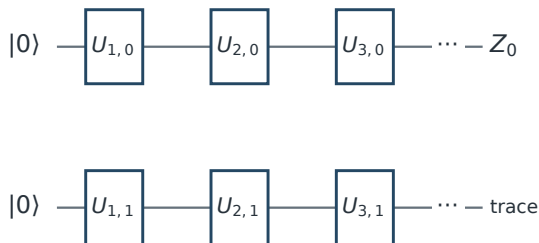
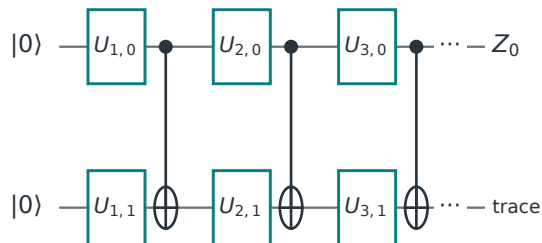
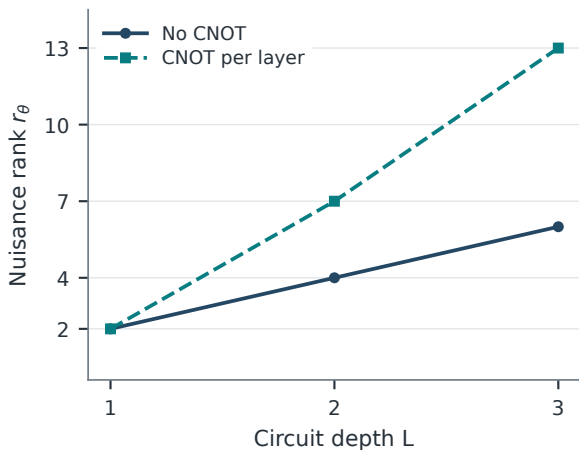


(a) Two qubits, no entangling gate

$$U_{\ell,q} = R_z R_x R_y (\alpha_\ell X + b_{\ell,q})$$

**(b) Same uploads, CNOT per layer**

$$U_{\ell,q} = R_z R_x R_y (\alpha_\ell X + b_{\ell,q})$$

**(c) Effective nuisance dimension****(d) Orders after nuisance profiling**